

Byte-enable memory

A thirty-two-bit word RAM can update one byte at a time using byte-enable bits

be equals 0001 starter

- Starter: four words by thirty-two bits
- Word zero is hex DDCCBBAA, bytes AA, BB, CC, DD little-endian
- Set addr zero
- Step clock: only byte zero changes from AA to fifty-five
- Word zero becomes DDCCBB55; BB, CC, DD unchanged
- Try be equals one-one-one-one for a full word, or zero-zero-one-one for a low halfword

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Byte-enable memory

Partial-word writes: we plus `be[3:0]` update only selected bytes of a 32-bit word — leave the others untouched.

Starter example: word 0 is DDCCBBAA — write with `be=0001` to change only the LSB.

Load starter example

Challenges (1/22)

Quiz: **be**. Partial-word write mask is called? Answer: byte enable

Answer

Show hint

Check

Next

Idle

quiz-be

quiz-lanes

quiz-lsb

quiz-zero

starter

toggle-be

be-lsb

be-full

step

patch-lsb

demo

half

half-step

zero-be

read

odd-lanes

explain

code-be0

we-off

reset

partial-flag

full-path

Core ideas

Word + lanes

32-bit word = 4 byte lanes; address selects the word.

`be[3:0]`

Each bit enables one lane on write.

Merge

Disabled lanes keep old data — RMW without a full read.

Byte-enable memory starter

learn_digital — byte enable mem

Workbook practice

- On paper, draw word zero with four byte lanes
- Mark be equals zero-zero-zero-one and din byte zero equals fifty-five
- Show the word before and after partial write
- Tabulate be patterns
- Name one pitfall: confusing be bit order with big-endian byte numbering

Pitfalls to watch

- Do not assume we equals one always writes the full word, check be
- Disabled lanes are not zeroed; they retain old values
- Little-endian means be zero is the least significant byte
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run the starter step and confirm only byte zero patches
- On paper, sketch one partial write with be equals zero-zero-one-one
- When you are ready, take the short quiz, then continue to async FIFO Gray pointers